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Methods for Genetic Study of Man

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Significance

- One of the reasons why Human Genetics is not highly developed as a discipline is because man is highly unfavorable for genetic study because of
  - **Social hindrances** render it impossible for controlled breeding experiments
  - **Number of offspring** of each marriage is also small, rendering statistical generalizations and calculations impossible
  - Since **man's lifespan is long**, the results cannot be studied for three or four generations
  - Majority of human are **genetically heterozygous** for many characters. Therefore, it is difficult to get **isogenic strains for genetic experiments**
- Hence methods to pursue an "INDIRECT" study

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Pedigree Analysis

- A method to understand the pattern of inheritance in man
- Because man has long life span, and little number of children, we cannot go forward to understand the inheritance
- Pedigree analysis is a method of going back in the generations to understand the inheritance of traits in man
- A pedigree chart is made after collecting information about the existing members of the family and as much information as possible on the previous generations

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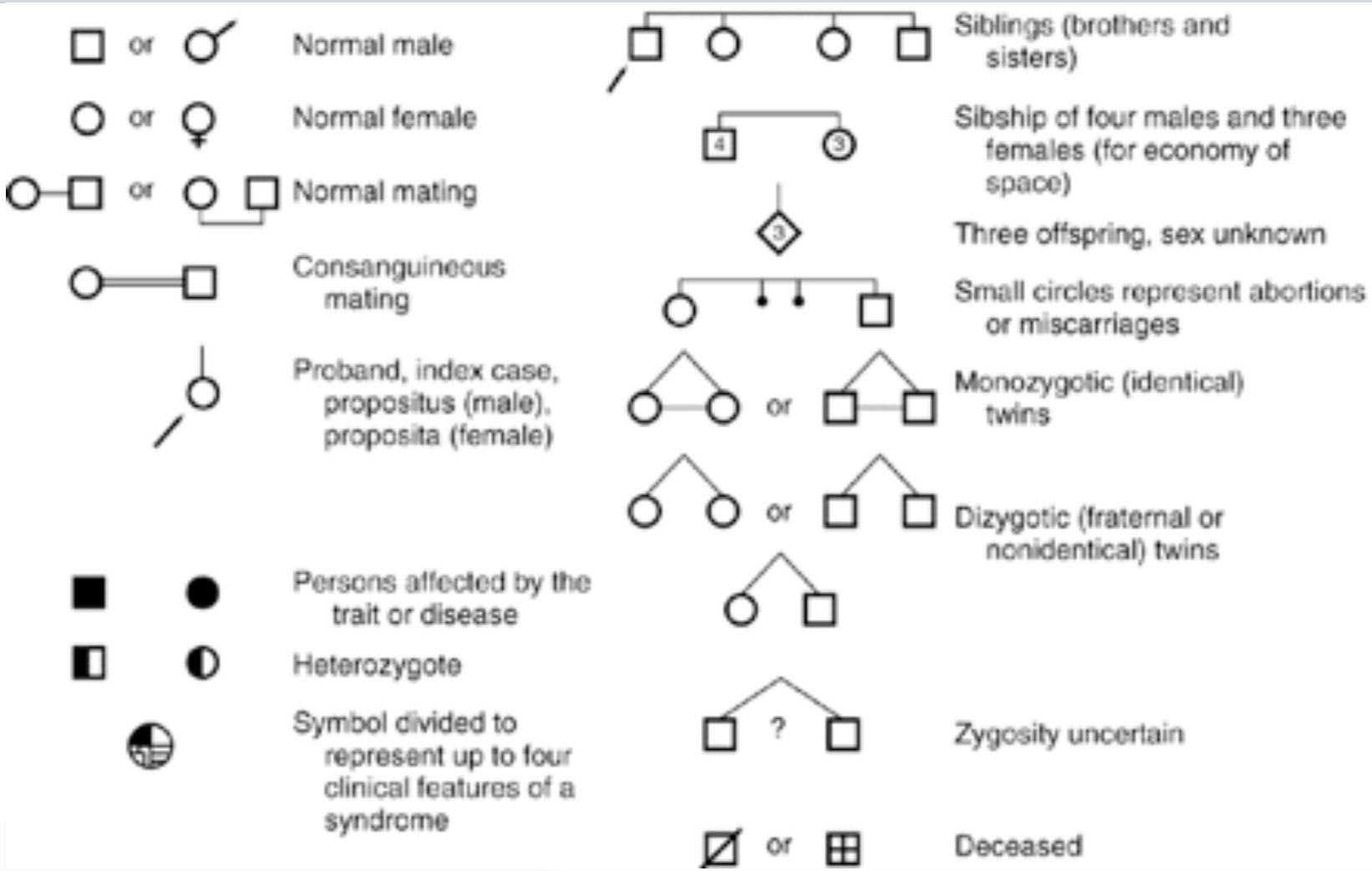
Pedigree Analysis

- Through pedigrees analysis, we can know whether a trait is
  - **Autosomal or Sex Linked**
  - **Dominant or Recessive**
- **Pedigrees are based on phenotype and not genotype.**
- For genotype, molecular studies are required

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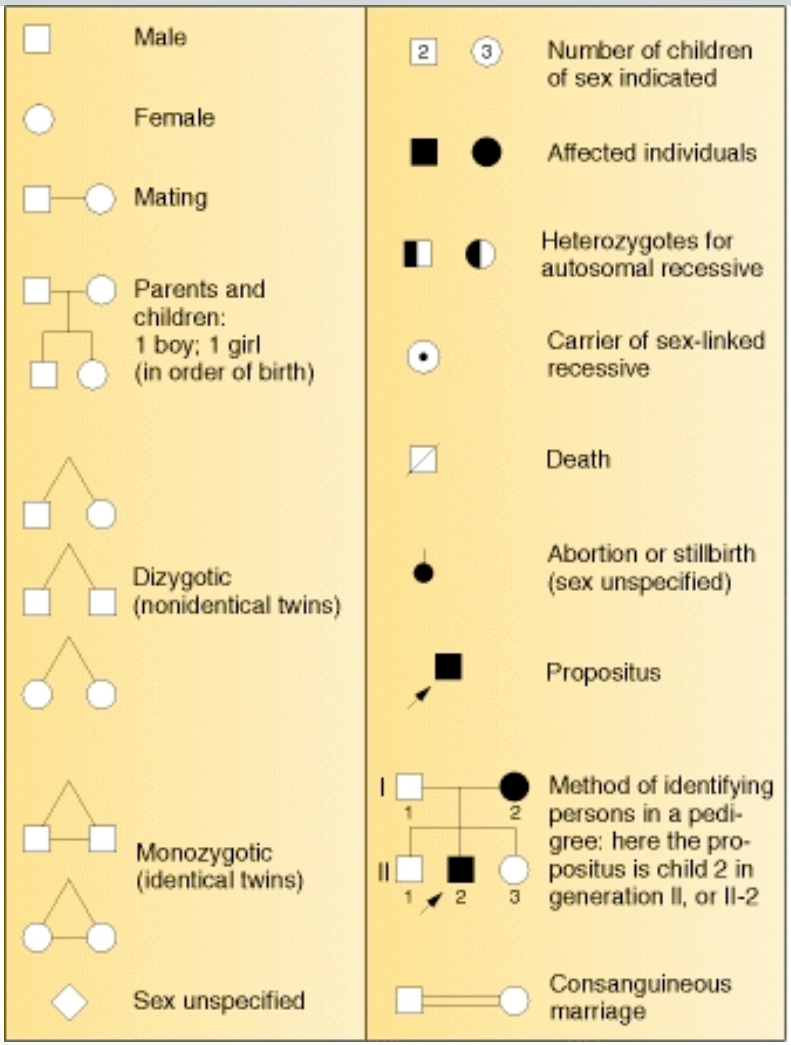
Pedigree Notation



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Pedigree Notation



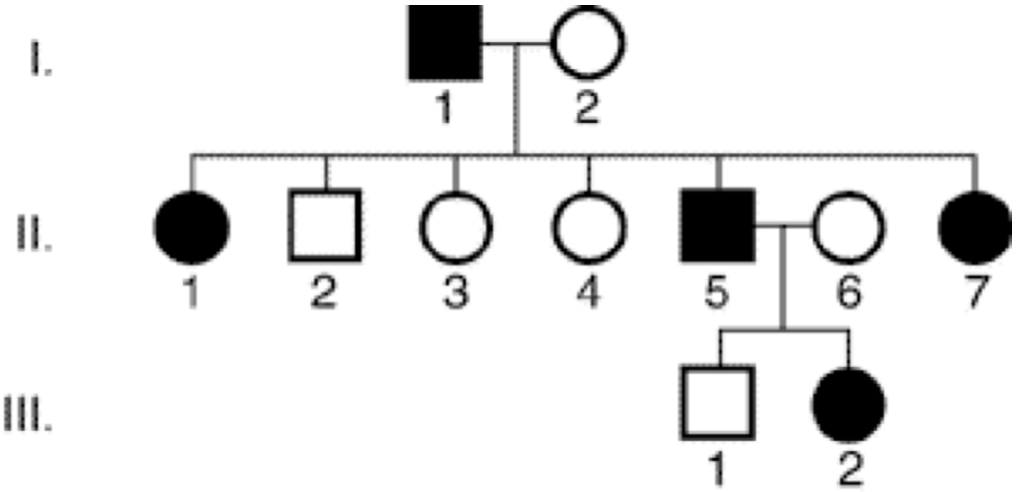
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Autosomal Dominance

Dominant – Not Recessive

- Roughly equal number of affected and unaffected persons
- Every affected person has an affected parent
- If one generation does not express the trait, it is lost forever
- Achondroplastic Dwarfism



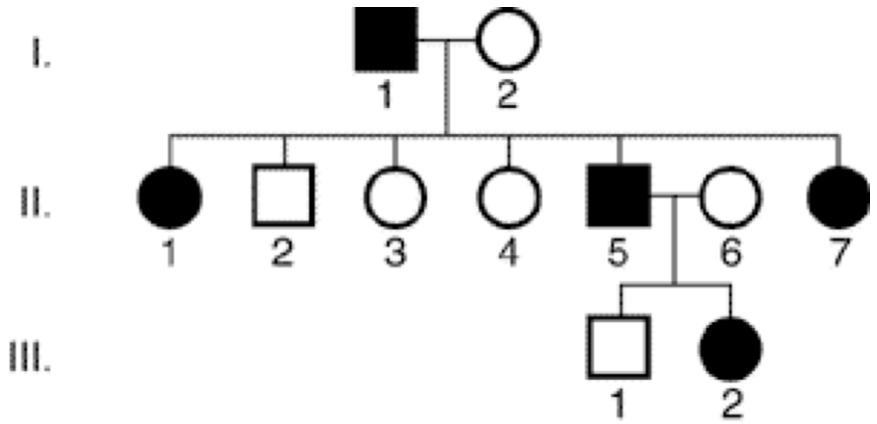
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Autosomal Dominance

Autosomal – Not Sex-linked

- Equal number of men and women got affected
- The affected man (1) passes it to his son – Hence not X-linked
- The affected man (1)passes it to his daughter – Hence not Y-linked



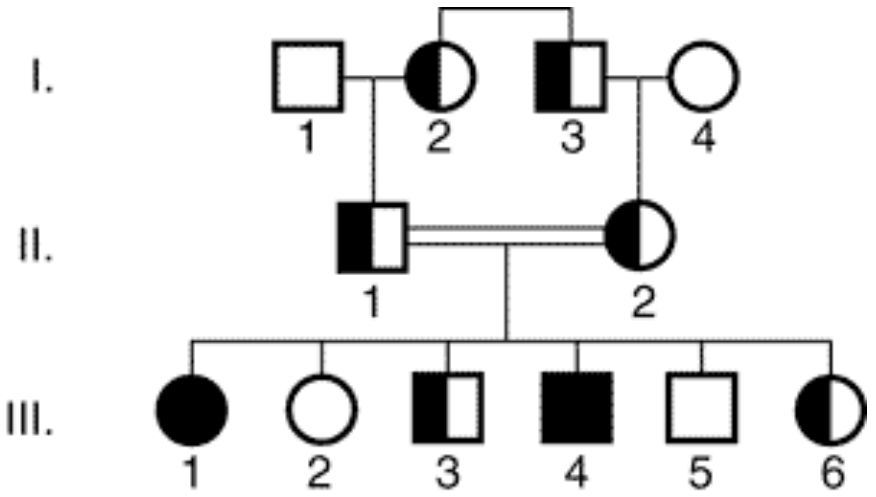
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Autosomal Recessive

Why Recessive?

- Normal parents with affected children, parents are heterozygous
- On an average ¼ children will be affected, ½ will be heterozygotes and ¼ will be normal
- All children of an affected person will be phenotypcially normal heterozygoes

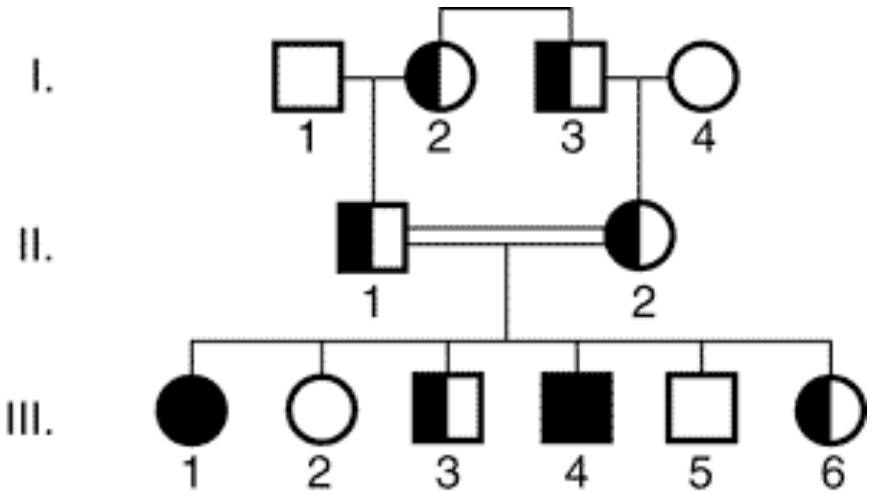


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Autosomal Recessive

- All children of two affected persons will be affected
- Skipping of generations due to masking of phenotypes
- Result of consanguineous marriages
- Ex. Sickle Cell Anemia, Albinism

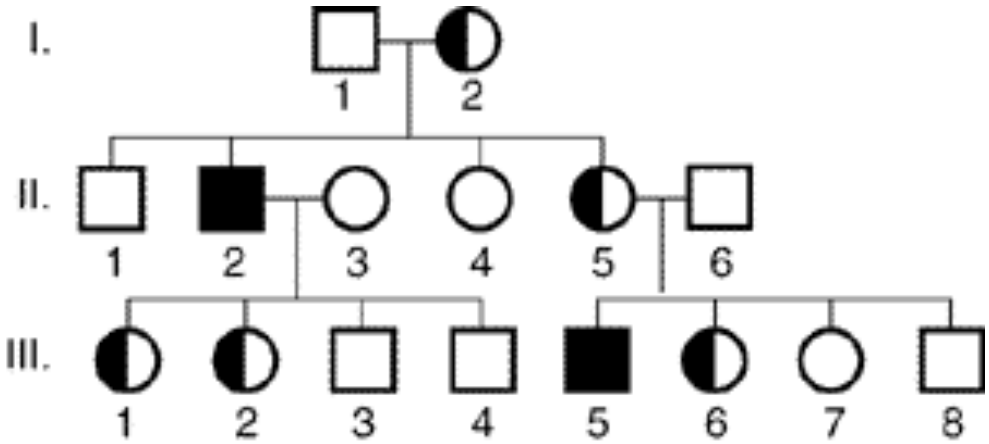


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X-Linked Recessive

- Nearly all affected persons are male.
- If the trait is transmitted through the heterozygous mother, she is usually phenotypically normal
- An affected male never transmits the trait to his sons.
- All daughters of an affected male will be carriers.
- The carrier female transmits the trait to 1/2 of her sons.



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Biochemical Method

- Every activity in the human body including the appearance of a phenotype is a biochemical activity
- Every biochemical activity is controlled by an enzymes
- The synthesis of an enzyme depends on the genes – protein synthesis
- This relation between Gene and enzyme is given by Beadle and Tatum and is called the **One Gene One Enzyme Hypothesis**

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Biochemical Method

- A number of biosynthetic steps transform a precursor to end product
- All the steps constitute a metabolic pathway
- Each step of biosynthetic pathway is catalyzed by an enzyme which in turn is synthesized by the control of a specific gene
- **Biochemical genetics** determines the activities of genes within cells to analyze the substrates and products of gene-controlled reactions

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Biochemical Genetics

- Biochemical genetics is widely used to identify inborn errors of metabolisms
- Many discovered – Alkaptonuria, Gout, PKU, Cystinuria, Galactosemia etc.
- A number of special techniques like **chromatography and electrophoresis, gel filtration** are used by these specialists to extract proteins and study their amino acid sequences and any disturbances in the polypedtide chains
- Any correlation to specific gene action can diagnosed and any corrective action can be taken, if possible

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Immunological Method

- Many proteins are antigenic – i.e., when introduced into an organism, they stimulate the production of antibodies
- Various antigens exist in blood
- All these antigens are genetically determined and their study constitutes **immunogenetics**
- Blood antigens of man include inherited variations and the particular combination of antigens in an individual is almost as unique as fingerprints

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Immunological Method

- Immunological techniques are used in blood group determinations, in organ transplants, Rh Factor compatibility etc.
- Some specific HLAs are associated with human diseases and disease predispositions
- Major area of research also includes **autoimmune diseases** like Diabetes Type 1, Multiple Sclerosis, Rheumatoid Arthritis etc.
- Immunological methods can also be used to understand genetic similarities & differences between individuals and between species (evolutionary distance) – anti-sera testing

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Twin Method

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What is Twin Method?

SOURCES OF VARIATION

- Genetic Variation
- Environmental Variation
- Interaction b/w Genetics and Environment

- Twin method is a mechanism through geneticists can ascertain whether a particular genetic trait has a strong genetic influence in its expression or a strong environmental influence
- What it calculates is **Heritability**

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Heritability Estimates

- Heritability is the proportion of variation that can be attributed to genetic factors

(H = Vg/Vg+Ve = Vg/Vt)

- An estimate of degree to which a particular character is genetically influenced – **Heritability Estimate**
- There are two types of twin methods to estimate heritability
  1. H/E Based on Variance between MZ and DZ Twins
  2. H/E Based on MZ Twins Reared Together and Apart

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How are Twins Formed

In 1874, **Camille Dereste** gave distinction between two types of twins – **Monozygotic Twins** and **Dizygotic Twins**

Monozygotic Twins

- They originate from a single zygote
- Zygote splits in the early state and develops into two ***genetically identical individuals*** because they both split from same zygote
- Hence, the MZ twins are also called **Identical Twins** and are of **same sex**

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How are Twins Formed

Dizygotic Twins

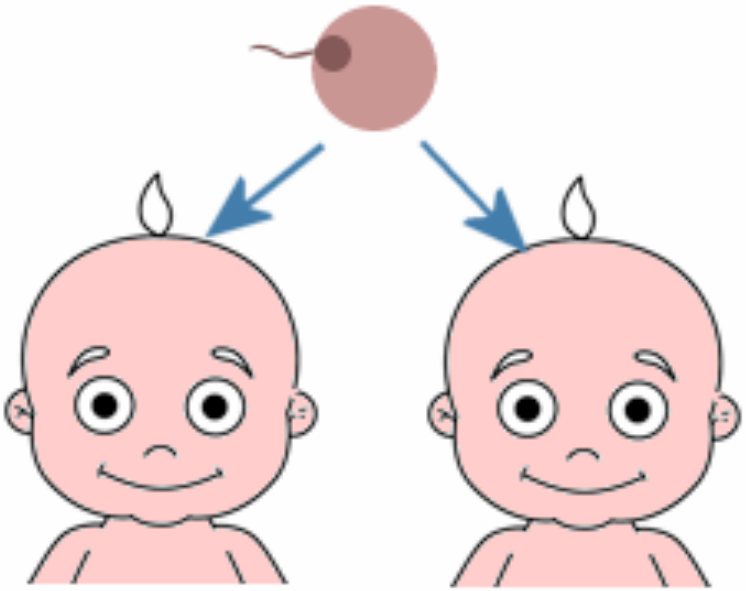
- Result from fertilization of two eggs by two sperms
- The relations between DZ twins are no different form that of **siblings**
- Hence, DZ twins are also called **Fraternal Twins** and may be of **different sex**

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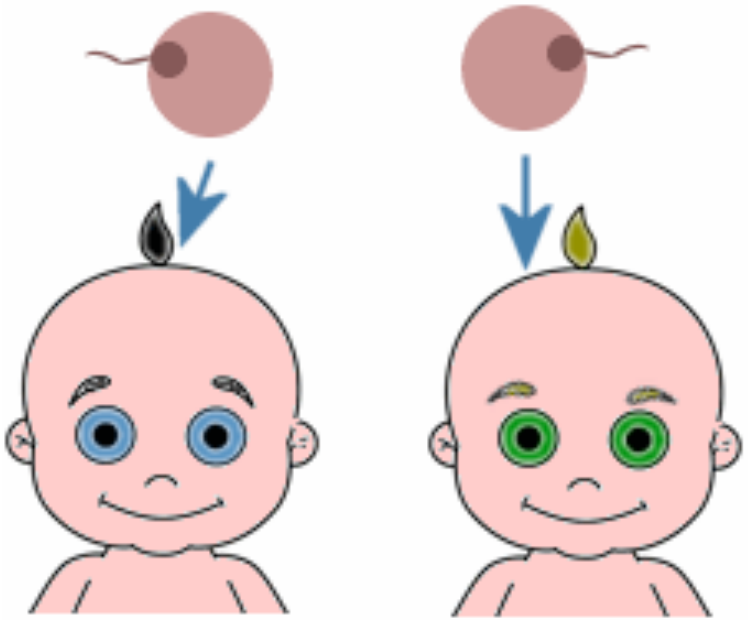
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Types of Twins

Monozygotic



Dizygotic



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Diagnosis of Zygosity

- Implies the diagnosis of a twin pair either as MZ or DZ
- It is the method to know whether the twin pair is born from same zygote or different zygote
- The diagnosis is extremely significant when the twins are of same sex and similar looks
- **Three Methods**
  - Placental Method
  - Similarity Method
  - DNA Fingerprinting

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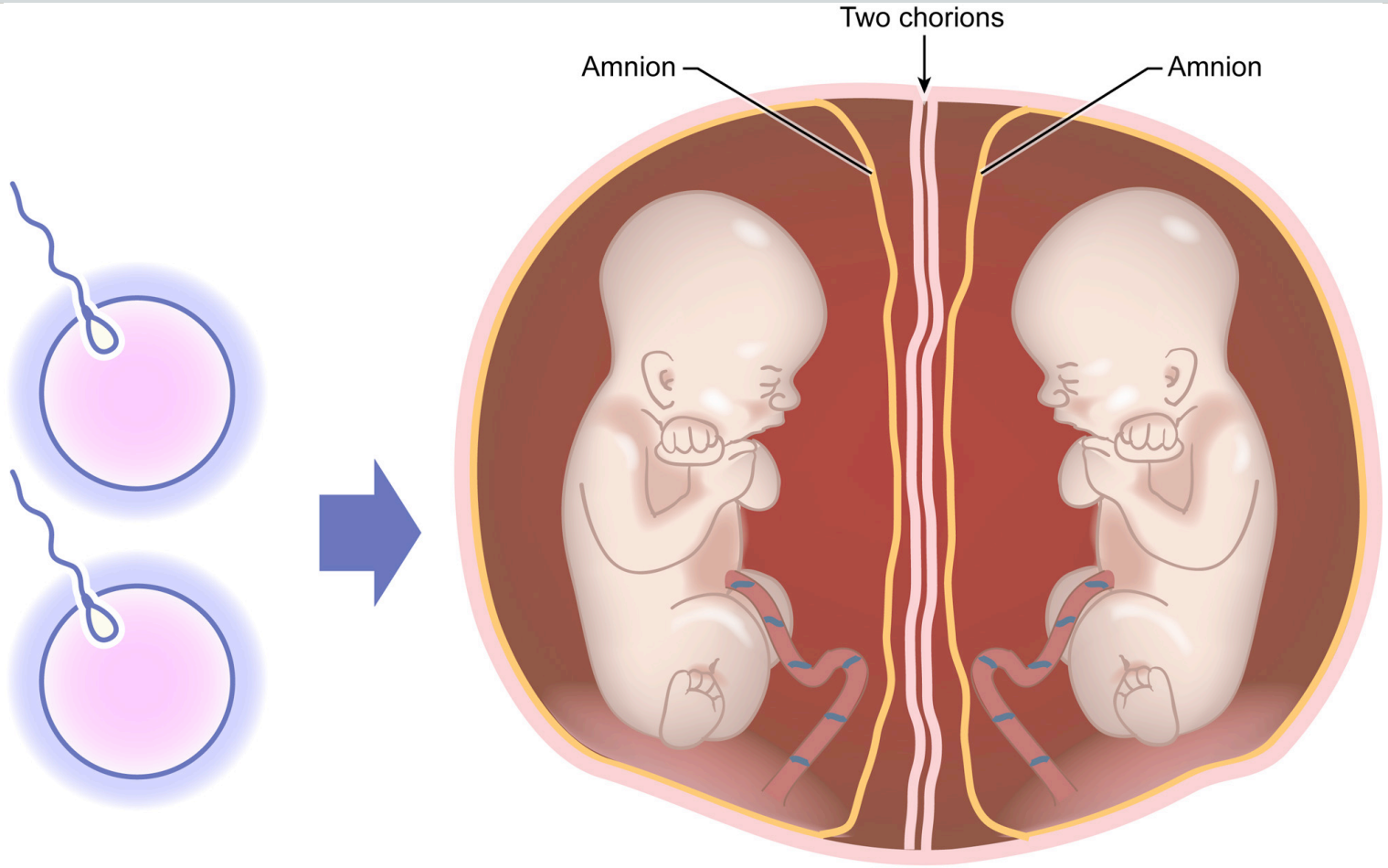
Placental Method

- Absolute diagnosis is difficult
- In the uterus of the mother, the developing zygote is surrounded by three layers
  - **Inner Amnion**
  - **Middle Chorion**
  - **Outer Placenta**
- Placental conditions vary in DZ and MZ twins

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Dizygotic Twins

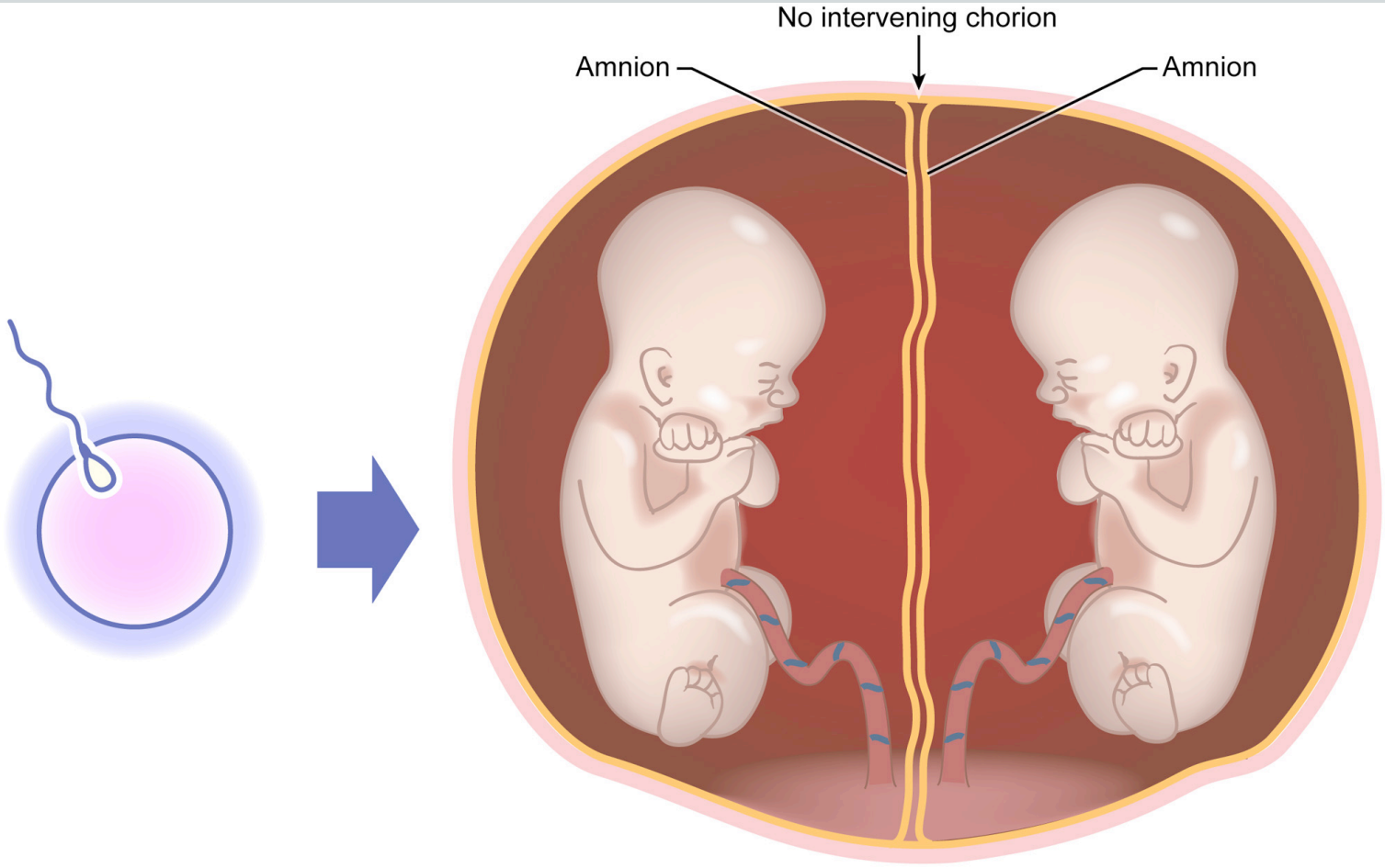


Dizygotic
(Dichorionic, Diamniotic)

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Monozygotic Twins

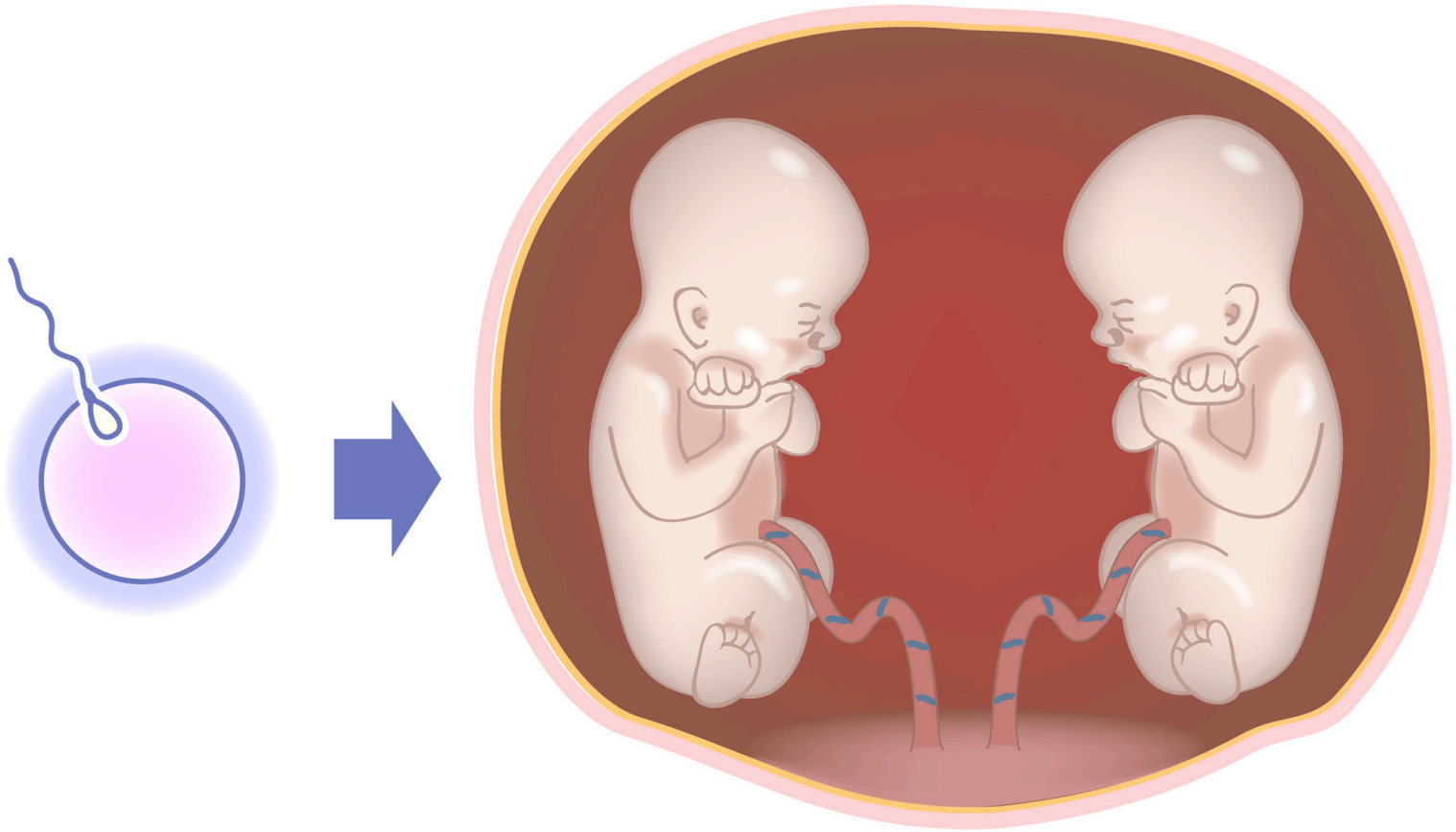


Monozygotic
(Monochorionic, Diamniotic)

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Monozygotic Twins



Monozygotic
(Monochorionic, Monoamniotic)

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Diagnosis of Zygosity

Similarity Method

- In this method, geneticists use traits which show individual variations like tissue antibodies.
- In order to diagnose the zygosity a small skin graft is made
- In this case, only MZ twin can accept the skin from its partner

DNA Fingerprinting– DNA Mapping between twins

No single method by itself will be authentic diagnosis

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Heredity & Environment

MZ TWINS

- Arising from same zygote – **Genetic Homogeneity**
- Any observed differences in their phenotypes – **purely environmental in origin**

DZ TWINS

- DZ twins – by contrast – differ because of both genetics and environment for a character

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Concordance Vs. Discordance

- **CONCORDANCE -** If both the members of the twin pair possess a character or both are free from it
- **DISCORDANCE -** If only one member possesses a trait
- The extent of concordance or discordance is a measure of relative roles of heredity and environment
- **Example:** If 100 pairs of each MZ and DZ twins are evaluated for a particular trait, 60 pairs of MZ twin show concordance compared to 30 pairs of DZ twins, does this have a strong genetic or environmental element???
- The trait can be said to have strong genetic element

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H/E based on variance between MZ & DZ twins

- This method uses the comparison of MZ and DZ twins for concordance and discordance
- Concordance and discordance rate indicates within-the-pair variation for MZ and DZ twins
- Once the within-the-pair variation is known, **Coefficient of Correlation** can be calculated thus…

Coefficient of Correlation = $(V_{DZ} - V_{MZ}) / V_{DZ}$

- V_{DZ} = Within-the-pair variation of DZ twins
- V_{MZ} = Within-the-pair variation of MZ twins

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H/E based on variance between MZ & DZ twins

- For a character with strong genetic tendency, the value of V_{MZ} will be extremely low, compared to value of V_{DZ}
- Hence, the value of Coefficient of Correlation will be reaching ONE
- If a particular character has strong environmental influence, the value of V_{MZ} would be high and may be same as the value of V_{DZ} – coefficient of correlation – approaches ZERO
- By the values of genetic and environmental component of all characters, we can calculate heritability using the formula

$$H = V_G / (V_G + V_E) = V_G / V_T$$

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H/E based on MZ reared together and reared apart

- The magnitude of variance between MZ twins reared together and reared apart are calculated
- This is based on the assumption that since the twins are genetically similar, any variation among them is purely environmental in susceptibility

COEFFICIENT OF CORRELATION

$$(V_A - V_T) / V_T$$

- V_A = Variation in MZ twins reared apart
- V_T = Variation in MZ twins reared together

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H/E based on MZ reared together and reared apart

$$(V_A - V_T) / V_T$$

- In case of strong environmental influence, the value of V_A would be greater and value of V_T would be small and hence the coefficient of correlation would be approaching ONE
- Similarly, if there is a strong genetic influence, the value of V_T and V_A would be similar and hence the value approaches ZERO

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Viability of Twin Method

1. Assumes presence of genetic and environmental components without actually specifying them – nature, locality and behavior of genes and physical, chemical and biological components of environment
2. Cannot be used in estimating complex characters like intelligence, because they are caused not either by heredity nor environment but by complex interaction b/w them
3. Assumes that MZ twins are exactly similar, but evidence shows that they too differ in characters like size and vigor
4. Cannot be used in estimating the intra-uterine environment – difference here might lead to low concordance in MZ twins

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Viability of Twin Method

5. Sampling of twins – need not be the environment of population as a whole
6. Inter population variations – not accounted for – Ex. A particular trait in one population may differ due to environment but it can differ due to genetics in another population
7. Twinning is a rare phenomenon – generalizations – questionable in their objectivity

To overcome some of the issues, **Co-Twin Method** was developed as a correction to Twin Method

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Foster Child Method

- Used in the analysis of nature-nurture aspects, especially for analysis of intelligence
- **Technique**
  1. Children are selected at random and are placed in different homes classified as good, average and poor
  2. Since the group of children selected is random, the genetic factor of the trait selected is assumed to be equally distributed in them
  3. After lapse of time, they are tested on different intelligence scales
  4. If intelligence has environmental component, children placed in good homes should score better than those in average and poor ones

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Foster Child Method

- Method – not free from biases and subjectivity

REQUIREMENTS - OSBORNE – 1951

1. Foster children must be placed in the adoptive homes quite early – uninfluenced by environmental of original home
2. No selective placement
3. Adequate samples of children from various social levels
4. Must be from one population to eliminate variations due to ethnicity and race

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Recombinant DNA Technology

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The Background

- The entire genetic code is like a language
  - **Letters:** Nucleotide Bases
  - **Words:** Codons
  - **Sentences:** Genes
  - **Language:** DNA
- Codons give instructions which specify and order amino acids
- The sequence of nucleotide bases to specify amino acids is universal – same elephant or a tree
- But these amino acid "words" can be arranged in innumerable ways to make thousands of proteins, both enzymes and structural proteins

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The Background

- Some of these "words" are actually punctuation marks that start or stop protein
- The group of codes arranged in a "grammatically" correct sequence constitute a gene
- Through complicated biochemical processes, the information contained in these genes are translated into what the organisms are actually made of
- Now the question – is it possible to "edit" this genetic text to get some desirable result?
- In 1970, biochemists at Stanford University showed that the genetic could indeed be transferred from one organism to another!

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rDNA Technology

- The terms "recombinant DNA technology," "DNA cloning," "molecular cloning,"or "gene cloning" all refer to the same process: the transfer of a DNA fragment of interest from one organism to a self-replicating genetic element such as a bacterial plasmid.
- **What is rDNA?** It is the DNA that has been cut from one strand of DNA(from one organism) and then inserted into the gap of another piece of DNA that has been broken (in another organism)

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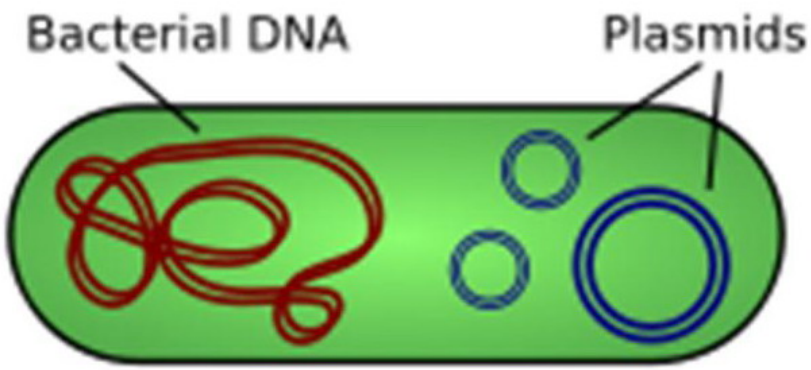
rDNA Technology

- The host DNA is often a bacterial cell such as *E coli*.
- The purpose of "splicing" the gene into the host DNA is to produce many copies of it.
- As bacteria reproduce in a very short time, it is possible to make millions of copies of the gene fairly quickly.

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rDNA Technology

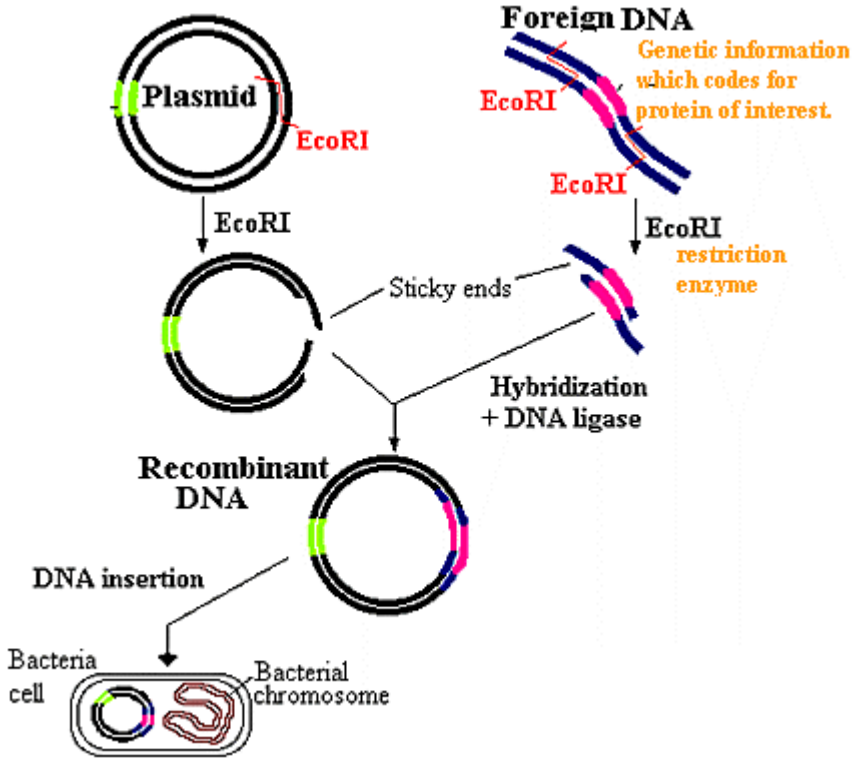


- Bacteria are often used in biotechnology as they have plasmids
- A plasmid is a circular piece of DNA that exists apart from the chromosome and replicates independently of it.

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rDNA Technology



1. Cut the Bacterial DNA with "restriction enzymes".
2. Remove bacterial DNA (plasmid).
3. Cut the DNA from another organism with "restriction enzymes" .
4. Combine the cut pieces of DNA together with another enzyme DNA Ligase and insert them into bacteria.
5. Reproduce the recombinant bacteria.
6. The foreign genes will be expressed in the bacteria.

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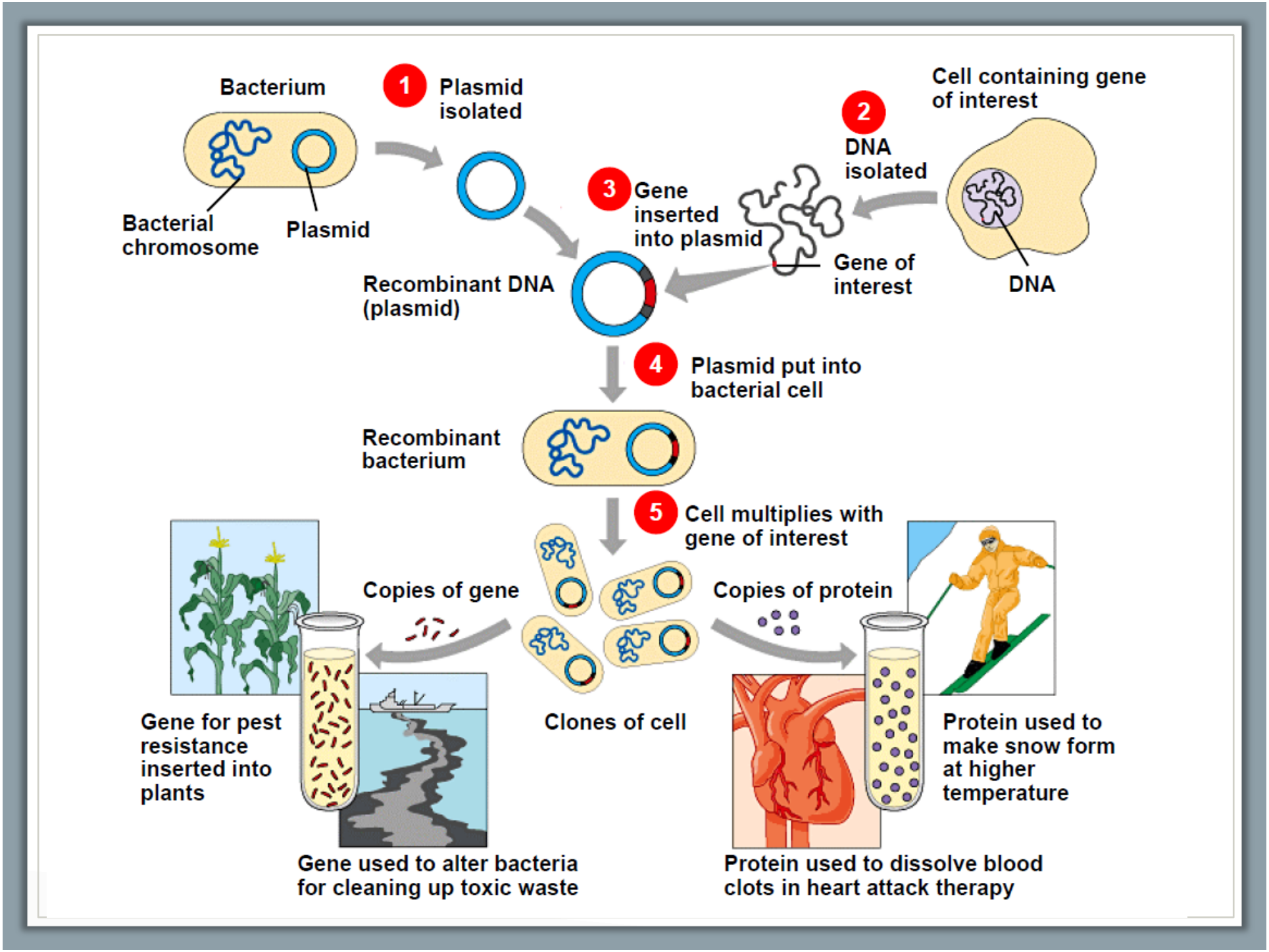
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rDNA Technology

- The recombinant DNA needs to be reintroduced into the bacteria so they can multiply and make more of the gene.
- Can be done by combining them in a test tube with $CaCl_2$.
- The high concentration of calcium ions makes the membranes of the bacteria more porous.
- This then allows the plasmids to move into the bacterial cells.
- In addition to bacteria, viral vectors are also use to inject the rDNA into animals and

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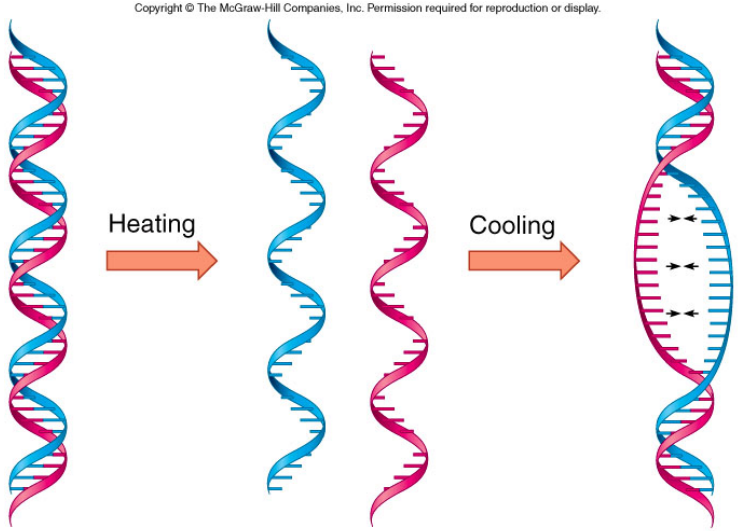
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Polymerase Chain Reaction

- PCR is a mechanism to make multiple copies of DNA from a very small sample
- The complementary strands of DNA can be separated and re-associated by heating and cooling
- PCR amplifies DNA which can be used in diagnosis and therapies

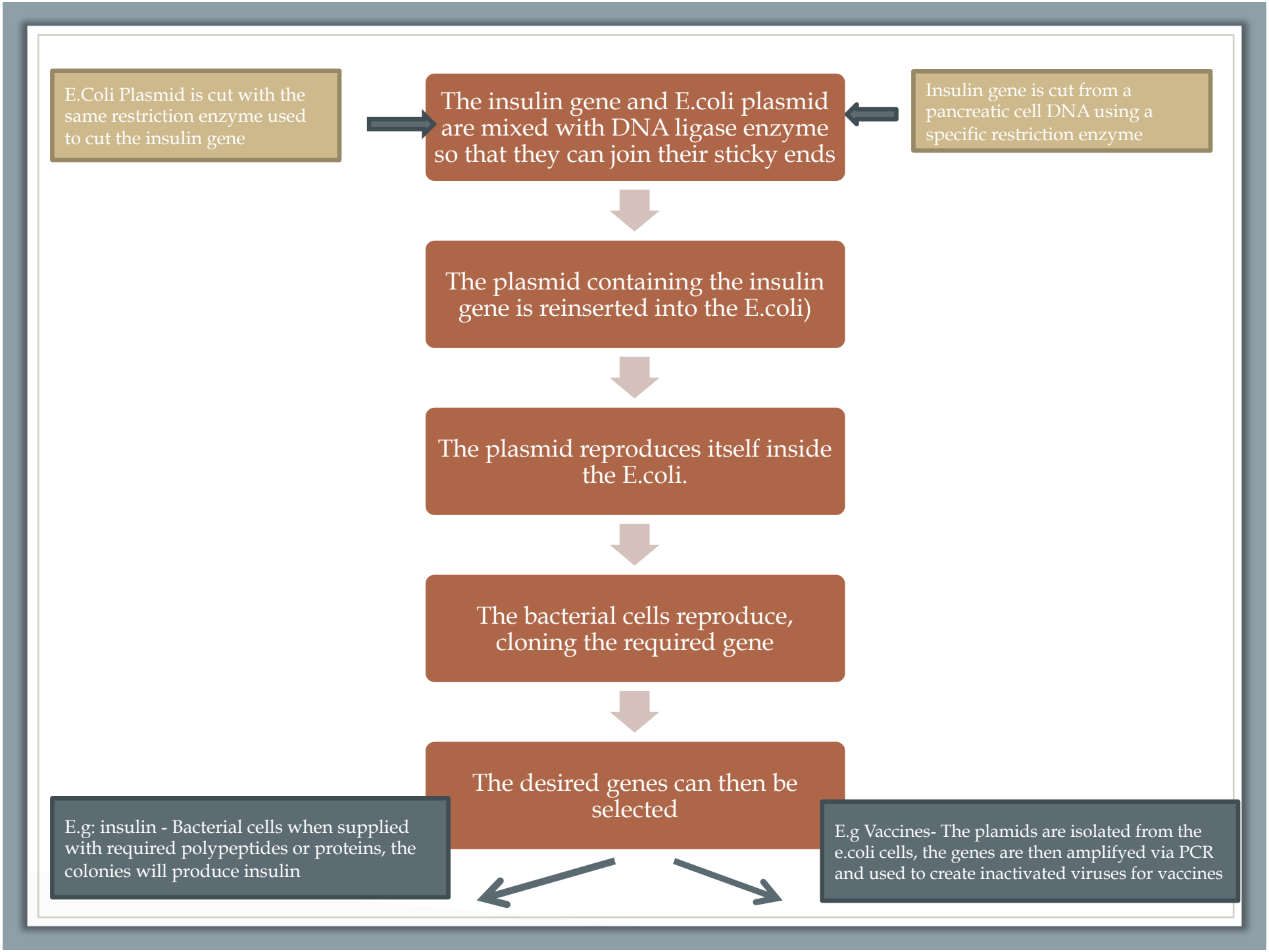
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(a) DNA heating and cooling. DNA responds to heat by denaturing—losing its hydrogen bonding, and thereby separating into its two strands. When cooled, the two strands rejoin at complementary sites. The two strands need not be from the same organisms as long as they have matching sites.

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Uses of rDNA Technology

- In addition to producing transgenic organisms like disease resistant plants, high yielding food crops, oil eating bacteria …
- rDNA technology has enormous potential in human diseases and medicine
  1. DNA Probes
  2. Gene Therapy
  3. Production of Hormones
  4. Production of Synthetic Vaccines

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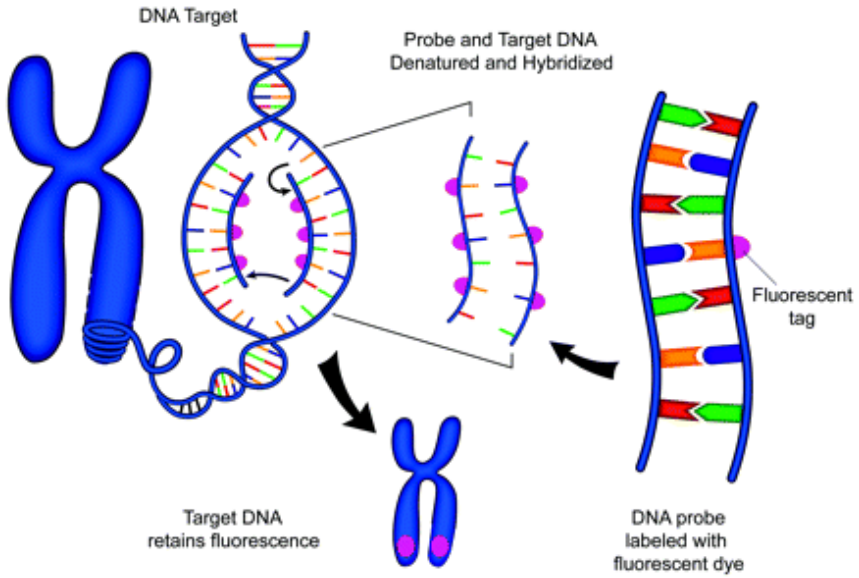
DNA Probes

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DNA Probes

- It is a radio actively labeled DNA with fluorescent tags which can be used to locate similar / complementary DNA in the genome of any organism
- Useful in diagnosis, DNA probes can hybridize with with specific DNA sequences and help diagnose
  - Specific parasitic DNA
  - Mutations at specific locations in genomes etc.



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DNA Probes

- Earlier identification of parasites required culture of blood or urine samples in different media - time consuming and dangers of contamination and false results
- Diagnosis of viruses using conventional methods of pathology was extremely difficult and DNA probes have now been made for a number of these parasites
- DNA probes have also been made for most of gene disorders / mutations enabling easy diagnosis of these diseases
- DNA probes have been developed for Leishmania, Trypanosoma, Plasmodium, Schistosoma, Wuchereria and other human parasites
- DNA probes are also used in DNA Profiling (will discuss)

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